

CE 201: Surveying: Traditional and Digital

Course Code	CE 201
Course Name	Surveying: Traditional and Digital
L-T-P-C	2-0-2-3
Prerequisites	None
Intended for	UG
Distribution	Discipline Core
Approval	9th Senate (May 2016)

Course Contents

- **Introduction to surveying:**
Understanding and need, Reconnaissance survey, compass survey, linear measurements. [2 Lectures]
- **Module II**
Type of maps, scales and uses, coordinate and map projection, plotting accuracy, toposheet numbering. [4 Lectures]
- **Module III**
Surveying equipment, levels, compass, theodolites, tachometer, EDM, total Stations and other instruments. [16 Lectures]
 - Levelling and contouring: types of instruments and their classifications, types and application of different methods, cross sections and gradient calculations.
 - Theodolites: Types and classification of different theodolites, applications
 - Total Station: Types, classification & applications
 - GPS & DGPS
- **Module IV**
Measurement of Distance, Direction and Elevation. [3 Lectures]
- **Module V**
Theory of error, adjustment of triangulation nets and level nets. Tachometry, Triangulation. [3 Lectures]

Laboratory Modules

1. Chain & tape Surveying: Distance measurements
2. Measurements of area using chain and tape.
3. Leveling: Profiling using Theodolite.
4. Leveling: Trigonometric using Theodolite.
5. Measurement of Area using Theodolite
6. Error estimation and calculations.
7. Total Station: Basic and Settings
8. Measurement of Area using Total Station
9. Profiling using Total Station.
10. GPS and DGPS survey and its basics and settings
11. Measurement of area and profiling using DGPS

Survey Camp: Reconnaissance and establishing the stations; Base line measurements, Triangulation readings on various stations; computation and preparation of triangulation map; contouring; preparation of map; preparation of report.

Geology Camp: Reconnaissance of the area; Elementary geological field mapping of rock formations and structural details; Geomorphic processes Preparation of report.

Text Books

1. B. C. Punmia, A.K. Jain and A.K. Jain, **Surveying, Vol-I and Vol-II**, Laxmi Publication Pvt., 1996.
2. G. W. Schofield, **Engineering Surveying**, Butterworth, Heinemann, New Delhi, 6th Ed., 2007.
3. Arora, K.R., **Surveying, Vol. I, II and III**, Standard Book House, 1995.
4. T. P. Kanetkar and S. V. Kulkarni, **Surveying and Levelling, Vol-I and Vol-II**, Pune Vidyarthi Griha Prakshan, 1972.
5. Leick, A., **GPS Satellite Surveying**, John Wiley, 2004.

References

1. R.N. Colwell (Editor-in-Chief), **Manual of Remote Sensing, Vol. I & II**, American Society of Photogrammetry, Falls Church, Virginia, 1983.
2. Anderson, J.M. and Mikhail, E.M., **Surveying: Theory and Practice**, McGraw Hill, 1998.
3. Kaplan, E.D. and Hegarty, C.J., **Understanding GPS: Principles and Applications**, Artech House, 2006.
4. Ahmed El-Rabbany, **Introduction to GPS: The Global Positioning System**, Artec House, London, 2002.
5. Jay Farrell, **Aided Navigation: GPS with High Rate Sensors: GPS with High Rate Sensors**, McGraw Hill, New York, 2008.